Offered at: San Luis Obispo Campus
The Plant Sciences Department at Cal Poly offers students an opportunity not just to learn, but to learn-by-doing. Our students benefit from a broad spectrum of opportunities ranging from hands-on experiences in our fields, groves, nurseries, and greenhouses to real world application through internships and other collaborations with our industry partners. We also excel in providing a foundational plant science background and instilling a passion for plants, as we produce the next generation of leaders in plant sciences.
Students in this major begin with core courses that provide a thorough introduction to the various concentrations. Each concentration, in turn, has required courses and electives, which may be shared by other concentrations. In their first year, students explore curricular and professional opportunities to enable them to choose a concentration. In consultation with professional and faculty advisors, students have the flexibility to select electives within the concentrations according to their career goals and interests.
Internships are readily available to students and are highly recommended. Interns are typically placed with private industry or public facilities across the United States but may also take place in foreign countries.
Program alumni are employed nationally and internationally and are often leaders in their industries. Many pursue careers in research and development or go on to attend graduate school in related fields. Graduates of the department are in great demand. Typically there are more internship and job opportunities than there are students and graduates to fill them.
Concentrations

+ Fruit and Crop Science

+ Environmental Horticultural Science

+ Plant Protection Science

Program Learning Objectives
Demonstrate technical competence in their concentration by identifying the majority of globally important food, and/or ornamental plants and demonstrating applications of theoretical sciences to their production, maintenance and post-harvest handling.
Effectively evaluate and adapt basic cultural practices, economic uses, and environmental interactions in the production of food, fiber, or ornamental plants.

Assess and implement appropriate sustainable growing and/or horticultural design practices based on region and microclimate, especially as they relate to water, soil and other natural resources.
Make informed and ethical decisions regarding environmental, social, and economic impacts of horticultural and agricultural activities and will contribute to their professions' continued relevancy by identifying, evaluating and responding to changing public perceptions, governmental regulations and industry challenges.
Practice a range of complex problem-solving exercises and excel in diagnosing and resolving plant health issues in outdoor and enclosed plant production systems.
Organize, synthesize, evaluate, and reconfigure information about complex, multivariate, living systems to gain new insights and communicate their findings to multiple stakeholder groups clearly, scientifically, and ethically.
Degree Requirements and Curriculum
In addition to the program requirements listed on this page, students must also satisfy requirements outlined in more detail in the Minimum Requirements for Graduation section of this catalog, including:
40 units of upper-division courses
2.0 GPA
Graduation Writing Requirements (GWR)
U.S. Cultural Pluralism (USCP)
Note: No Major, Support or Concentration courses may be selected as credit/no credit. In addition, no more than 12 units of cooperative or internship courses can count towards your degree requirements.

MAJOR COURSES		
PLSC 1101	Orientation to Plant Sciences	1
PLSC 1120 & 1120L	Principles of Plant Sciences and Principles of Plant Sciences Lab	3
PLSC 1124	Plant Propagation	3
PLSC 3304	Introduction to Plant Breeding	3
PLSC 3313	Agricultural Entomology	3
PLSC 3321	Weed Biology and Management	4
PLSC 3323	Plant Pathology	3
PLSC 3351	Experimental Techniques and Analysis	3
PLSC 4410	Crop Physiology	3
PLSC 4461	Senior Project I	1
PLSC 4462	Senior Project II	1
Concentration		
(See list of Concentrations below)		37
SUPPORT COURSES		
BOT 1121	General Botany (5B & 5C) [1]	4

BRAE 3340	Irrigation Water Management (Upper-Division 2/5) [1]	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) [1]	4
MATH 1006	College Algebra (2) [1]	3
SS 1120	Introductory Soil Science	4
SS 2221	Soil Health and Plant Nutrition	4
STAT 1110	Applied Statistical Concepts and Methods	3
GENERAL EDUCATION (GE)		
(See GE program requirements below)		30
FREE ELECTIVES		
Free Electives		0
Total Units		**120**

[1] Required in Major or Support; also satisfies General Education (GE) requirement.

Concentrations

+ Fruit and Crop Science
+ Environmental Horticultural Science
+ Plant Protection Science

General Education (GE) Requirements

+ General Education (GE) Requirements

+ Plant Sciences
+ Fruit and Crop Science

- Environmental Horticultural Science

Environmental Horticultural Science Concentration - BS in Plant Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence. Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

TERM 1		UNITS
PLSC 1101	Orientation to Plant Sciences	1
PLSC 1120 & 1120L	Principles of Plant Sciences and Principles of Plant Sciences Lab	3

BOT 1121	General Botany (5B & 5C) [1]	4
MATH 1006	College Algebra (2) [1]	3
General Education Requirement (1A)		3
	Units	**14**

TERM 2

PLSC 1124	Plant Propagation	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) [1]	4
General Education Requirement (1B)		3
General Education Requirement (1C)		3
	Units	**13**

Second Year

TERM 1

PLSC 1123	Introduction to Sustainable Site Horticulture	3
PLSC 2234	Introduction to Plant Materials	3
SS 1120	Introductory Soil Science	4
Approved Elective		3
General Education Requirement		3
	Units	**16**

TERM 2

PLSC 3301	Horticultural Production Techniques	3
PLSC 3321	Weed Biology and Management	4
SS 2221	Soil Health and Plant Nutrition	4
STAT 1110	Applied Statistical Concepts and Methods	3
General Education Requirement		3
	Units	**17**

Third Year

TERM 1

PLSC 3313	Agricultural Entomology	3
PLSC 3323	Plant Pathology	3
PLSC 3351	Experimental Techniques and Analysis	3
BRAE 3340	Irrigation Water Management (Upper-Division 2/5) [1]	3

	Units	**12**
TERM 2		
PLSC 3304	Introduction to Plant Breeding	3
PLSC 3332	Sustainable Site Design and Systems	3
PLSC 3334	Advanced Plant Materials	3
PLSC 3340	Principles of Greenhouse Environments	3
General Education Requirement		3
	Units	**15**
Fourth Year		
TERM 1		
PLSC 4461	Senior Project I	1
Approved Elective		3
Approved Elective		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
	Units	**16**
TERM 2		
PLSC 4410	Crop Physiology	3
PLSC 4427	Disease and Pest Control Systems for Ornamental Plants	4
PLSC 4462	Senior Project II	1
Approved Elective		3
Approved Elective		3
General Education Requirement		3
	Units	**17**
	Total Units	**120**

[1] Required in Major or Support; also satisfies General Education (GE) requirement.

- Plant Protection Science

Plant Protection Science Concentration - BS in Plant Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

TERM 1		UNITS
PLSC 1101	Orientation to Plant Sciences	1
PLSC 1120 & 1120L	Principles of Plant Sciences and Principles of Plant Sciences Lab	3
BOT 1121	General Botany (5B & 5C) [1]	4
MATH 1006	College Algebra (2) [1]	3
General Education Requirement (1A)		3
	Units	**14**
TERM 2		
PLSC 1124	Plant Propagation	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) [1]	4
General Education Requirement (1B)		3
General Education Requirement (1C)		3
	Units	**13**

Second Year

TERM 1		
SS 1120	Introductory Soil Science	4
Approved Elective		3
Approved Elective		3
Approved Elective		3
General Education Requirement		3
	Units	**16**
TERM 2		
PLSC 3321	Weed Biology and Management	4
SS 2221	Soil Health and Plant Nutrition	4
STAT 1110	Applied Statistical Concepts and Methods	3
Approved Elective		3
General Education Requirement		3
	Units	**17**

Third Year

TERM 1

PLSC 3313	Agricultural Entomology	3
PLSC 3323	Plant Pathology	3
PLSC 3351	Experimental Techniques and Analysis	3
BRAE 3340	Irrigation Water Management (Upper-Division 2/5) [1]	3
	Units	**12**

TERM 2

PLSC 3304	Introduction to Plant Breeding	3
PLSC 4406	Advanced Weed Management	4
PLSC 4427	Disease and Pest Control Systems for Ornamental Plants	4
Approved Elective		3
General Education Requirement		3
	Units	**17**

Fourth Year

TERM 1

PLSC 2205 or PLSC 2212 or PLSC 3333	Orchard and Vegetable Enterprise Project or Environmental Horticulture Enterprise Project or Greenhouse Vegetable Production	2
PLSC 4428	Advanced Plant Pathology	3
PLSC 4441	Biological Control for Pest Management	3
PLSC 4461	Senior Project I	1
General Education Requirement		3
General Education Requirement		3
	Units	**15**

TERM 2

PLSC 4410	Crop Physiology	3
PLSC 4431	Integrated Pest Management for Insects	3
PLSC 4462	Senior Project II	1
Approved Elective		3
General Education Requirement		3

General Education Requirement	3
Units	**16**
Total Units	**120**

[1] Required in Major or Support; also satisfies General Education (GE) requirement.

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Biomedical EngineeringToggle Biomedical Engineering
Civil & Environmental EngineeringToggle Civil & Environmental Engineering
Computer EngineeringToggle Computer Engineering
Computer Science and Software EngineeringToggle Computer Science and Software Engineering
Electrical EngineeringToggle Electrical Engineering
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General EngineeringToggle General Engineering
Industrial & Manufacturing EngineeringToggle Industrial & Manufacturing Engineering
Materials EngineeringToggle Materials Engineering
Mechanical EngineeringToggle Mechanical Engineering
College of Liberal ArtsToggle College of Liberal Arts
Art & DesignToggle Art & Design
Communication StudiesToggle Communication Studies
EnglishToggle English
Ethnic StudiesToggle Ethnic Studies
Graphic CommunicationToggle Graphic Communication
HistoryToggle History
Interdisciplinary Studies in the Liberal ArtsToggle Interdisciplinary Studies in the Liberal Arts
JournalismToggle Journalism
MusicToggle Music
PhilosophyToggle Philosophy
Political ScienceToggle Political Science
Psychology and Child DevelopmentToggle Psychology and Child Development
Social SciencesToggle Social Sciences
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